# Percent Change from Old Price to New Price

## Spelled out methods...

Remember, use decimal form of percentages in calculations. For example, an increase of 23% means `percent_increase` is 0.23 and the `scale_factor` is 1.23.

### Find `new_price` from `old_price` and `percent_increase`

`scale_factor` = 1 + `percent_increase`

`new_price` = `old_price` * `scale_factor`

### Find `new_price` from `old_price` and `percent_decrease`

`scale_factor` = 1 - `percent_decrease`

`new_price` = `old_price` * `scale_factor`

### Find `old_price` from `new_price` and `percent_increase`

`scale_factor` = 1 + `percent_increase`

`old_price` = `new_price` / `scale_factor`

### Find `old_price` from `new_price` and `percent_decrease`

`scale_factor` = 1 - `percent_decrease`

`old_price` = `new_price` / `scale_factor`

### Find `percent_increase` from `old_price` and `new_price`

`scale_factor` = `new_price` / `old_price`

`percent_increase` = `scale_factor` - 1

### Find `percent_decrease` from `old_price` and `new_price`

`scale_factor` = `new_price` / `old_price`

`percent_decrease` = 1 - `scale_factor`

## Formulas with variables

$$\begin{aligned} s &= \text{scale factor} \\ i &= \text{initial (old) price} \\ n &= \text{new price} \\ c &= \text{percent change (decimal form)} \end{aligned}$$

$$s = \frac{n}{i}$$

$$i = \frac{n}{s}$$

$$n = si$$

$$c = |s - 1|$$

- If new price is **more** than initial price, the percent change is a percent **increase**, and $s = c + 1$, and $c = s - 1$, and:

$$c = \frac{n}{i} - 1$$

$$i = \frac{n}{1 + c}$$

$$n = i \cdot (1 + c)$$

- If new price is **less** than initial price, the percent change is a percent **decrease**, and $s = 1 - c$, and $c = 1 - s$, and:

$$c = 1 - \frac{n}{i}$$

$$i = \frac{n}{1 - c}$$

$$n = i \cdot (1 - c)$$